

The AD-AS Model

EC201

Today we'll go beyond the IS-LM model to think about supply and what happens in the long run (when prices are not fixed).

- ① Aggregate Demand (AD) curve.
- ② Short run and long run Aggregate Supply (AS) curve.
- ③ The AD-AS model.

- In the IS-LM model aggregate demand is the main determinant of the final output. This is why there is no supply in the IS-LM model.
- Before Keynes, supply-side analysis was king.
- Say's Law: "Supply creates its own demand".
- We may assume that Say's law applies in the long run (when prices are flexible), but in the short run (sticky prices) some of the Keynesian features from the IS-LM model are valid.
- The AD-AS model was developed to include analysis of supply and demand.

Q: How are output and aggregate prices determined (in the short run and the long run)?

Upward sloping AS curve

Long run:

- Assume the classical case holds.
- ⇒ The long run aggregate supply curve (LRAS) is vertical.

Short run:

- In the short run we want to consider a positively sloped SRAS curve.
- There are 3 main economic models to explain why the SRAS is positively sloped:
 - ① Sticky Wages
 - ② Sticky Prices
 - ③ The Lucas' Imperfect Information Model

- Output: $Y = F(K, L)$.
- In the short run nominal wages (W) are sticky (e.g. fixed wage contract).
- Labour markets are competitive: $W/P = MPL$ and $L^D = g(W/P)$ with $g' < 0$.
- What matters for firms and workers is the real wage ($\frac{W}{P}$).
- An increase in the price level leads to a decrease in the real wage, which causes firms to hire more workers and produce more output.

Sticky wages (2)

Firms and workers bargain over the nominal wage before knowing the future price level (P^e).

$$W = \omega \times P^e$$

Where ω is the real wage target such that labor demand equals labor supply (full-employment $\Rightarrow$ firms produce at the natural level $\bar{Y}$).

$$W = \omega \times P^e$$

- If $P < P^e \Rightarrow$ Realized real wage greater than target $\Rightarrow$ Firm hires fewer workers $\Rightarrow Y < \bar{Y}$
- If $P > P^e \Rightarrow$ Realized real wage less than target $\Rightarrow$ Firm hires more workers $\Rightarrow Y > \bar{Y}$
- If $P = P^e \Rightarrow$ Realized real wage equals target $\Rightarrow Y = \bar{Y}$

Sticky wages (4)

Implied SRAS curve:

$$Y = \bar{Y} + \alpha(P - P^e)$$

where $\alpha > 0$.

Normally written as:

$$P = P^e + \frac{1}{\alpha}(Y - \bar{Y}) \quad (\text{Lucas Aggregate Supply})$$

⇒ SRAS is a relationship between output and **unexpected movements** in the aggregate price level.

The difference $(Y - \bar{Y})$ is the **output gap**

Sticky prices

- Market structure: **monopolistic competition** $\Rightarrow$ firms are price-setters.
- Some firms change prices quickly (flexible) while others set prices in advance (sticky).
- Desired price depends on:
 - Aggregate price level: $P \uparrow \Rightarrow$ Costs $\uparrow \Rightarrow$ Higher price.
 - Aggregate output: $Y \uparrow \Rightarrow$ Demand $\uparrow \Rightarrow$ Higher price.

Can express this as:

$$p = P + a(Y - \bar{Y}), \quad \text{where } a > 0.$$

Sticky prices (2)

- **Flexible price firms:**

$$p = P + a(Y - \bar{Y})$$

- **Sticky price firms** form expectations:

$$p = P^e + a(Y^e - \bar{Y})$$

- Suppose $Y^e = \bar{Y}$ and proportion of sticky firms is s , flexible is $(1 - s)$.

$$\text{Aggregate price: } P = sP^e + (1 - s)P + (1 - s)a(Y - \bar{Y})$$

Sticky prices (3)

After manipulation:

$$P = P^e + \frac{1-s}{s}a(Y - \bar{Y})$$

⇒ SRAS determined by unexpected changes in demand and proportion of sticky-price firms.

Lucas' Imperfect Information Model

- Firms know their own price, but not competitors' prices (uncertainty about overall price level).
- All wages and prices are fully flexible.
- If firms observe demand increase, they don't know if it's specific to them or economy-wide.

$$y(z) = \bar{y}(z) + \gamma(p(z) - P)$$

where $y(z)$ = quantity by firm z , $p(z)$ = price by firm z , P = aggregate price.

- Increase in demand for firm z only: $p(z) \uparrow$ relative to $P \Rightarrow y(z) \uparrow$
- Increase in total demand: both $p(z)$ and $P \uparrow \Rightarrow y(z)$ unchanged

Lucas' Imperfect Information Model (2)

Firms must form expectations:

$$y(z) = \bar{y}(z) + \gamma(p(z) - P^e)$$

- Firms place probability q on aggregate demand increase, $(1 - q)$ on firm-specific increase.
- Firms' expectations generally incorrect $\Rightarrow$ over/underestimate price level $\Rightarrow y(z)$ varies.
- Aggregating all firms:

$$Y = \bar{Y} + \gamma(P - P^e)$$

Aggregate Supply (AS)

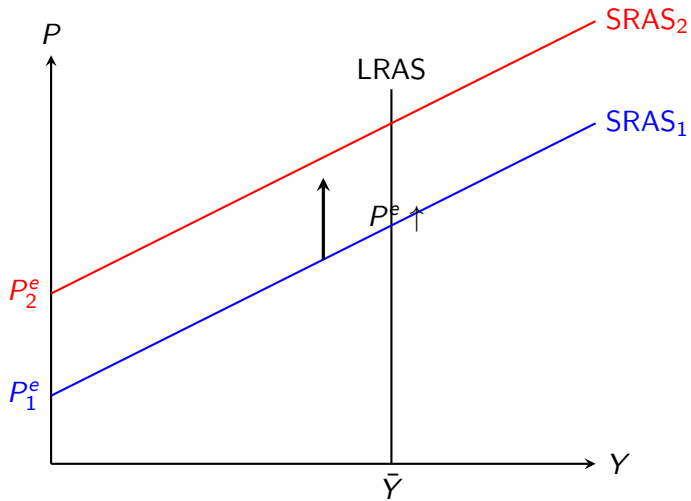
Given $\beta > 0$ we can write the short-run aggregate supply curve as:

$$P = P^e + \beta(Y - \bar{Y})$$

The slope of the aggregate supply:

- Sticky wage model: $\beta = 1/\alpha$
- Sticky price model: $\beta = a((1-s)/s)$
- Lucas's imperfect information model: $\beta = 1/\gamma$

The position of the AS curve



$P^e \uparrow$ Changes in expectations shift the AS curve

Expectations

- We have introduced price expectations P^e
- How do firms and individuals form expectations?
- Two main approaches in macroeconomics:
 - ① Adaptive Expectations
 - ② Rational Expectations

Adaptive Expectations

Agents base expectations on **past observations** only

$$P_{t+1,t}^e = P_{t,t-1}^e + \lambda(P_t - P_{t,t-1}^e)$$

- $\lambda \in (0, 1)$
- Systematic forecast errors: $P_t - P_{t,t-1}^e$
- If $\lambda = 1$ then $P_{t+1,t}^e = P_t$ (expect tomorrow = today)

Rational Expectations

To avoid systematic errors:

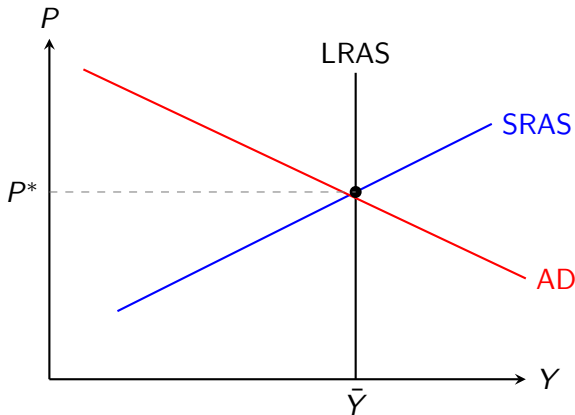
$$P_{t+1,t}^e = E(P_{t+1}|\Omega_t) = E_t(P_{t+1})$$

- E_t = mathematical expectation of P_{t+1} conditional on information set Ω_t
- Information set contains all relevant information.
- No systematic errors: $E_t(P_{t+1} - E_t P_{t+1}) = 0$

AS-AD model

Putting it together:

- Downward sloping AD curve (from IS-LM)
- Upward sloping SRAS curve (from sticky prices, sticky wages, or Lucas)
- Vertical LRAS curve



AS-AD model: Shocks

Two types of shocks:

1. Demand shocks (AD)

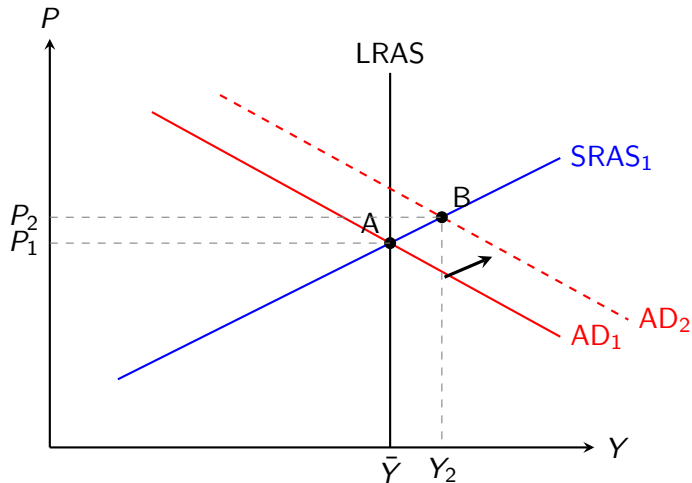
- **Positive:** AD shifts right (e.g. unexpected tax cut)
- **Negative:** AD shifts left (e.g. unexpected tax increase)

2. Supply shocks (AS)

- **Positive:** SRAS shifts right (e.g. unexpected cost decrease)
- **Negative:** SRAS shifts left (e.g. unexpected energy price shock)

AS-AD model: Positive demand shock

Short-run: Higher demand $\Rightarrow P$ and $Y \uparrow$



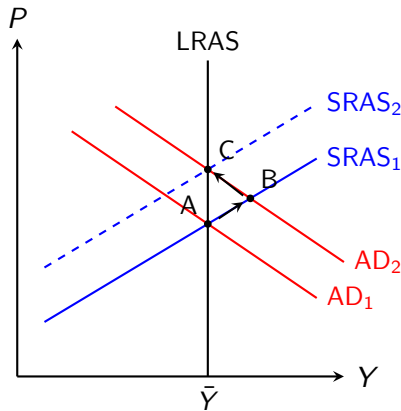
AS-AD model: Positive demand shock (2)

Short-run (A to B):

- AD shifts right
- $P \uparrow$ from P_1 to P_2
- $Y \uparrow$ above $\bar{Y}$

Long-run (B to C):

- Expectations adjust: $P^e \uparrow$
- SRAS shifts left
- Y returns to $\bar{Y}$
- $P \uparrow$ to P_3



AS-AD model: Economic policy

- In the AS-AD model we always return to natural output in the long run.
- But we don't know how long it will take.
- Economic policy can speed up the adjustment.

Example: Negative demand shock

- Short run: Recession (output below $\bar{Y}$, falling prices)
- Without policy: Wait for expectations to adjust $\Rightarrow$ slow
- With policy: Expansionary policy (increase G or M) $\Rightarrow$ faster return to $\bar{Y}$

AS-AD model: Policy response

$AD \downarrow$ can be offset by monetary/fiscal policy

